Structure of the variable self.HP containing results of the habit plane analysis

L. Heller

February 5, 2026

```
parent_cluster_id → dict [10 keys]
    child_cluster_id → dict [4 keys]
        parent_idx → int
        child_idx → int
        semi_roi_1 → dict [30 keys]
            G2Cryst_red → dict [2 keys]
                A → array[9×3×3] <float64>
                M → array[14×3×3] <float64>
            G2Cryst_red_avg → dict [2 keys]
                A → array[3×3] <float64>
                M → array[3×3] <float64>
            G2Sampl_red → dict [2 keys]
                A → array[9×3×3] <float64>
                M → array[14×3×3] <float64>
            G2Sampl_red_avg → dict [2 keys]
                A → array[3×3] <float64>
                M → array[3×3] <float64>
            <100>PF → dict [2 keys]
                A → array[3×27] <float64>
                M → array[3×42] <float64>
            <100>PF_avg → dict [2 keys]
                A → array[3×3] <float64>
                M → array[3×3] <float64>
            T_AM_ebsd → array[4×3×3] <float64>
            Closest Variant → int
            Closes equivalent G2Cryst → dict [1 keys]
                M → array[3×3] <float64>
            Closes equivalent G2Cryst_allvars → list [12 items]
                [0] → dict [1 keys]
                    M → array[3×3] <float64>
            Theory G2Cryst → dict [1 keys]
                M → array[3×3] <float64>
            Theory G2Cryst_allvars → list [12 items]
                [0] → dict [1 keys]
                    M → array[3×3] <float64>
            Theory <100>PF_avg → dict [1 keys]
                M → array[3×3] <float64>
```

```
Theory <100>PF_avg_allvars → list [12 items]
     [0] → dict [1 keys]
          M → array[3×3] <float64>
Theory <100>PF_avg_symop → dict [1 keys]
     M → array[3×3] <float64>
Theory <100>PF_avg_symop_allvars → list [12 items]
     [0] → dict [1 keys]
          M → array[3×3] <float64>
Martensite symop → array[3×3] <float64>
Martensite symop_allvars → list [12 items]
     [0] → array[3×3] <float64>
Closest Theory OR → array[3×3] <float64>
Closest Theory OR_allvars → list [12 items]
     [0] → array[3×3] <float64>
Exp OR → array[3×3] <float64>
Exp OR_allvars → list [12 items]
     [0] → array[3×3] <float64>
dR OR → Rotation
dR OR_allvars → list [12 items]
     [0] → Rotation
Misori OR → float
Misori OR_allvars → list [12 items]
     [0] → float
Transformation strain → list [3 items]
     [0] → dict [1 keys]
          along [1. 0. 0.] [-] → float
Transformation strain_allvars → list [12 items]
     [0] → list [3 items]
          [0] → dict [1 keys]
               along [1. 0. 0.] [-] → float
OR → dict [3 keys]
     OR:A-M → array[3×3] <float64>
     axis → array[3] <float64>
     angle → float
Interfaces → dict [10 keys]
     x → array[23] <float32>
     y → array[23] <float32>
     linfit → array[2] <float64>
     linfitnorm → array[2] <float64>
     interface_trace_sample → array[3] <float64>
     interfacenorm_trace_sample → array[3] <float64>
     interface_trace → dict [2 keys]
          A → array[3] <float64>
          M → array[3] <float64>
     interfacenorm_trace → dict [2 keys]
          A → array[3] <float64>
          M → array[3] <float64>
     HP_guess → dict [2 keys]
          A → dict [7 keys]
               n_vec → array[10×3] <float64>
```

```
                n_vec_sampl → array[10×3] <float64>
                n_miller → array[10×3] <float64>
                n_miller_family → array[10×3] <float64>
                n_miller_normvec → array[10×3] <float64>
                n_miller_normvec_sampl → array[10×3] <float64>
                HPvsTrace_angle → array[10] <float64>
            M → dict [7 keys]
                n_vec → array[15×3] <float64>
                n_vec_sampl → array[15×3] <float64>
                n_miller → array[15×3] <float64>
                n_miller_family → array[15×3] <float64>
                n_miller_normvec → array[15×3] <float64>
                n_miller_normvec_sampl → array[15×3] <float64>
                HPvsTrace_angle → array[15] <float64>
        HP_matches → dict [3 keys]
            Score → dict [5 keys]
                low index habit plane → list [3 items]
                    [0] → float
                mean misalignment → list [3 items]
                    [0] → float
                number of corresponding directions → list [3 items]
                    [0] → int
                overall → list [3 items]
                    [0] → float
                fitcorresp → list [3 items]
                    [0] → str: 'yes'
            A → dict [2 keys]
                Habit plane normal → dict [8 keys]
                    misalign → list [3 items]
                        [0] → float
                    n_vec → list [3 items]
                        [0] → array[3] <float64>
                    n_vec_sampl → list [3 items]
                        [0] → array[3] <float64>
                    n_miller → list [3 items]
                        [0] → array[3] <float64>
                    n_miller_family → list [3 items]
                        [0] → array[3] <float64>
                    n_miller_normvec → list [3 items]
                        [0] → array[3] <float64>
                    n_miller_normvec_sampl → list [3 items]
                        [0] → array[3] <float64>
                    HPvsTrace_angle → list [3 items]
                        [0] → float
                Habit plane direction → dict [8 keys]
                    misalign → list [3 items]
                        [0] → list [2 items]
                            [0] → float
                    n_vec → list [3 items]
                        [0] → array[2×3] <float64>
```

```
                    n_vec_sampl → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller_family → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller_normvec → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller_normvec_sampl → list [3 items]
                        [0] → array[2×3] <float64>
                    HPvsTrace_angle → list [3 items]
                        [0] → array[2] <float64>
            M → dict [2 keys]
                Habit plane normal → dict [8 keys]
                    misalign → list [3 items]
                        [0] → float
                    n_vec → list [3 items]
                        [0] → array[3] <float64>
                    n_vec_sampl → list [3 items]
                        [0] → array[3] <float64>
                    n_miller → list [3 items]
                        [0] → array[3] <float64>
                    n_miller_family → list [3 items]
                        [0] → array[3] <float64>
                    n_miller_normvec → list [3 items]
                        [0] → array[3] <float64>
                    n_miller_normvec_sampl → list [3 items]
                        [0] → array[3] <float64>
                    HPvsTrace_angle → list [3 items]
                        [0] → float
                Habit plane direction → dict [8 keys]
                    misalign → list [3 items]
                        [0] → list [2 items]
                            [0] → float
                    n_vec → list [3 items]
                        [0] → array[2×3] <float64>
                    n_vec_sampl → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller_family → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller_normvec → list [3 items]
                        [0] → array[2×3] <float64>
                    n_miller_normvec_sampl → list [3 items]
                        [0] → array[2×3] <float64>
                    HPvsTrace_angle → list [3 items]
                        [0] → array[2] <float64>
semi_roi_2 → dict [30 keys]
    G2Cryst_red → dict [2 keys]
```

```
     A → array[9×3×3] <float64>
     M → array[12×3×3] <float64>
G2Cryst_red_avg → dict [2 keys]
     A → array[3×3] <float64>
     M → array[3×3] <float64>
G2Sampl_red → dict [2 keys]
     A → array[9×3×3] <float64>
     M → array[12×3×3] <float64>
G2Sampl_red_avg → dict [2 keys]
     A → array[3×3] <float64>
     M → array[3×3] <float64>
<100>PF → dict [2 keys]
     A → array[3×27] <float64>
     M → array[3×36] <float64>
<100>PF_avg → dict [2 keys]
     A → array[3×3] <float64>
     M → array[3×3] <float64>
T_AM_ebsd → array[4×3×3] <float64>
Closest Variant → int
Closes equivalent G2Cryst → dict [1 keys]
     M → array[3×3] <float64>
Closes equivalent G2Cryst_allvars → list [12 items]
     [0] → dict [1 keys]
          M → array[3×3] <float64>
Theory G2Cryst → dict [1 keys]
     M → array[3×3] <float64>
Theory G2Cryst_allvars → list [12 items]
     [0] → dict [1 keys]
          M → array[3×3] <float64>
Theory <100>PF_avg → dict [1 keys]
     M → array[3×3] <float64>
Theory <100>PF_avg_allvars → list [12 items]
     [0] → dict [1 keys]
          M → array[3×3] <float64>
Theory <100>PF_avg_symop → dict [1 keys]
     M → array[3×3] <float64>
Theory <100>PF_avg_symop_allvars → list [12 items]
     [0] → dict [1 keys]
          M → array[3×3] <float64>
Martensite symop → array[3×3] <float64>
Martensite symop_allvars → list [12 items]
     [0] → array[3×3] <float64>
Closest Theory OR → array[3×3] <float64>
Closest Theory OR_allvars → list [12 items]
     [0] → array[3×3] <float64>
Exp OR → array[3×3] <float64>
Exp OR_allvars → list [12 items]
     [0] → array[3×3] <float64>
dR OR → Rotation
dR OR_allvars → list [12 items]
```

```
    [0] → Rotation
Misori OR → float
Misori OR_allvars → list [12 items]
    [0] → float
Transformation strain → list [3 items]
    [0] → dict [1 keys]
        along [1. 0. 0.] [-] → float
Transformation strain_allvars → list [12 items]
    [0] → list [3 items]
        [0] → dict [1 keys]
            along [1. 0. 0.] [-] → float
OR → dict [3 keys]
    OR:A-M → array[3×3] <float64>
    axis → array[3] <float64>
    angle → float
Interfaces → dict [10 keys]
    x → array[21] <float32>
    y → array[21] <float32>
    linfit → array[2] <float64>
    linfitnorm → array[2] <float64>
    interface_trace_sample → array[3] <float64>
    interfacenorm_trace_sample → array[3] <float64>
    interface_trace → dict [2 keys]
        A → array[3] <float64>
        M → array[3] <float64>
    interfacenorm_trace → dict [2 keys]
        A → array[3] <float64>
        M → array[3] <float64>
    HP_guess → dict [2 keys]
        A → dict [7 keys]
            n_vec → array[11×3] <float64>
            n_vec_sampl → array[11×3] <float64>
            n_miller → array[11×3] <float64>
            n_miller_family → array[11×3] <float64>
            n_miller_normvec → array[11×3] <float64>
            n_miller_normvec_sampl → array[11×3] <float64>
            HPvsTrace_angle → array[11] <float64>
        M → dict [7 keys]
            n_vec → array[15×3] <float64>
            n_vec_sampl → array[15×3] <float64>
            n_miller → array[15×3] <float64>
            n_miller_family → array[15×3] <float64>
            n_miller_normvec → array[15×3] <float64>
            n_miller_normvec_sampl → array[15×3] <float64>
            HPvsTrace_angle → array[15] <float64>
    HP_matches → dict [3 keys]
        Score → dict [5 keys]
            low index habit plane → list [4 items]
                [0] → float
            mean misalignment → list [4 items]
```

```
        [0] → float
    number of corresponding directions → list [4 items]
        [0] → int
    overall → list [4 items]
        [0] → float
    fitcorresp → list [4 items]
        [0] → str: 'yes'
A → dict [2 keys]
    Habit plane normal → dict [8 keys]
        misalign → list [4 items]
            [0] → float
        n_vec → list [4 items]
            [0] → array[3] <float64>
        n_vec_sampl → list [4 items]
            [0] → array[3] <float64>
        n_miller → list [4 items]
            [0] → array[3] <float64>
        n_miller_family → list [4 items]
            [0] → array[3] <float64>
        n_miller_normvec → list [4 items]
            [0] → array[3] <float64>
        n_miller_normvec_sampl → list [4 items]
            [0] → array[3] <float64>
        HPvsTrace_angle → list [4 items]
            [0] → float
    Habit plane direction → dict [8 keys]
        misalign → list [4 items]
            [0] → list [2 items]
                [0] → float
        n_vec → list [4 items]
            [0] → array[2×3] <float64>
        n_vec_sampl → list [4 items]
            [0] → array[2×3] <float64>
        n_miller → list [4 items]
            [0] → array[2×3] <float64>
        n_miller_family → list [4 items]
            [0] → array[2×3] <float64>
        n_miller_normvec → list [4 items]
            [0] → array[2×3] <float64>
        n_miller_normvec_sampl → list [4 items]
            [0] → array[2×3] <float64>
        HPvsTrace_angle → list [4 items]
            [0] → array[2] <float64>
M → dict [2 keys]
    Habit plane normal → dict [8 keys]
        misalign → list [4 items]
            [0] → float
        n_vec → list [4 items]
            [0] → array[3] <float64>
        n_vec_sampl → list [4 items]
```

```
          [0] → array[3] <float64>
     n_miller → list [4 items]
          [0] → array[3] <float64>
     n_miller_family → list [4 items]
          [0] → array[3] <float64>
     n_miller_normvec → list [4 items]
          [0] → array[3] <float64>
     n_miller_normvec_sampl → list [4 items]
          [0] → array[3] <float64>
     HPvsTrace_angle → list [4 items]
          [0] → float
Habit plane direction → dict [8 keys]
     misalign → list [4 items]
          [0] → list [2 items]
               [0] → float
     n_vec → list [4 items]
          [0] → array[2×3] <float64>
     n_vec_sampl → list [4 items]
          [0] → array[2×3] <float64>
     n_miller → list [4 items]
          [0] → array[2×3] <float64>
     n_miller_family → list [4 items]
          [0] → array[2×3] <float64>
     n_miller_normvec → list [4 items]
          [0] → array[2×3] <float64>
     n_miller_normvec_sampl → list [4 items]
          [0] → array[2×3] <float64>
     HPvsTrace_angle → list [4 items]
          [0] → array[2] <float64>
```